Deploy

📈 **ML StatArb**

Krauss et al. (2017) · IEOR 4733

Navigation

- Paper Summary
- Reproduction Results
- Differences vs Original
- Robustness Checks
- Regime Sensitivity
- Risk Diagnostics

CONTROLS

Portfolio size k

10

Transaction cost (bps/side)

5

No-lookahead: train ends before test starts
Seed=1 · Daily rebalancing · Equal-weight

Paper Summary

Krauss, Do & Huck (2017) — *Deep neural networks, gradient-boosted trees, random forests: Statistical arbitrage on the S&P 500*

UNIVERSE	SAMPLE PERIOD	OUR EXTENSION	FEATURES
S&P 500	**1995–2015**	**1992–2024**	**31 lags**
1,288 constituents	Krauss paper period	32 Study Periods	R1–R20 + R40–R240

METHODOLOGY

Research Question
Can ML models (DNN, GBT, RF) generate statistically significant long-short alpha on S&P 500 daily returns?

Target Variable Y
Binary: does stock i outperform the cross-sectional median return on day t+1?

Training Design

- Rolling window: 750 trading days train → 250 trading days test
- 32 non-overlapping Study Periods (1990–2024)
- Strict temporal separation — no lookahead bias

Portfolio Construction

- Rank stocks by $\hat{P}(Y=1)$ each day
- Long top-k, short bottom-k (equal-weighted, dollar-neutral)
- $k \in \{10, 50, 100, 150, 200\}$

MODELS & RESULTS

Model	Description	Paper Sharpe	Our Sharp
DNN	31-31-10-5-2 Maxout, Adadelta, L1=1e-5	2.44	-0.02
GBT	XGBoost: 100 trees, depth=3, lr=0.1	3.16	0.56 (avg)
RF	1000 trees, depth=20, √p features	3.09	-0.05
ENS1	Equal-weight DNN+GBT+RF	3.40	2.40

ENSEMBLE EQUATIONS

$$\hat{P}^{ENS1} = \frac{1}{M}\sum_i \hat{P}^i$$

$$\hat{P}^{ENS2} = \sum_i w^i \hat{P}^i, \quad w^i = \frac{g^i}{\sum_j g^j}$$

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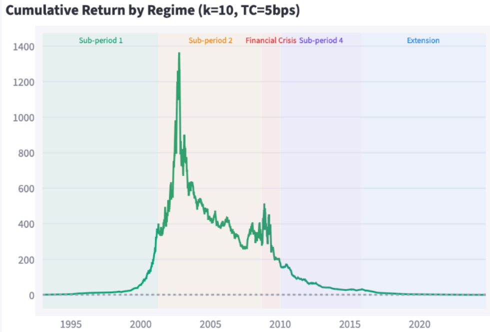
Regime Sensitivity

Performance across bull/bear, crisis, and macro-regime sub-periods

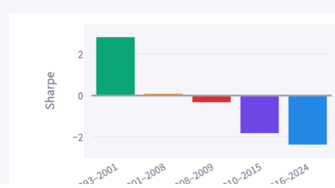
SUB-PERIOD 1 1993–2001	SUB-PERIOD 2 2001–2008	FINANCIAL CRISIS 2008–2009	SUB-PERIOD 4 2010–2015	EXTENSION 2016–2024
2.82	**0.08**	**-0.33**	**-1.83**	**-2.38**
Sharpe · 111% ann	Sharpe · 2% ann	Sharpe · -19% ann	Sharpe · -26% ann	Sharpe · -42% ann

Cumulative Return by Regime (k=10, TC=5bps)



REGIME PERFORMANCE TABLE

Regime	Ann Ret	Sharpe	Max DD
Sub-period 1 1993–2001	111.3%	2.82	-16.4%
Sub-period 2 2001–2008	2.5%	0.08	-81.2%
Financial Crisis 2008–2009	-19.5%	-0.33	-65.2%
Sub-period 4 2010–2015	-26.1%	-1.83	-85.7%
Extension 2016–2024	-41.6%	-2.38	-99.4%



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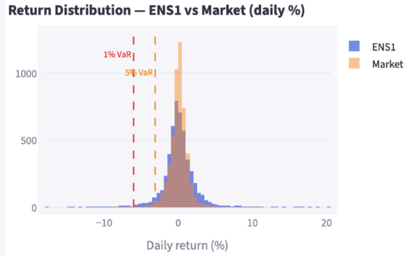
Risk Diagnostics

Factor exposure, tail risk & performance attribution — ENS1 k=10, Paper Period (1992–2015)

MAX DRAWDOWN	95% VAR	95% CVAR	SKEWNESS	EXCESS KURTOSIS
-93.1%	-3.14%	-5.14%	0.53	10.3
Sharpe: 0.51	Daily 5th pctile	Expected tail loss	Right > 0 preferred	Fat tails > 0

Drawdown (%) — ENS1 k=10, Paper Period

Return Distribution — ENS1 vs Market (daily %)



TAIL RISK — ENS1 VS MARKET

Metric	ENS1	Market	ENS1/Market
1% VaR	-0.0604	-0.0334	1.81x
1% CVaR	-0.0864	-0.0480	1.8x
5% VaR	-0.0314	-0.0190	1.65x

FACTOR REGRESSION (NEWEY-WEST SE)

Factor	FF3	FF3+Rev+Mom	FF5
Alpha (daily)	0.0027***	0.0027***	0.0027***
Market (Mkt-RF)	-0.0022	-0.0205	-0.0061
SMB	-0.0422	-0.0452	-0.0278